Contents

1 Introduction 3
2 Usage 3
2.1 Project installation 3
3 Reference 3
3.1 circuit 3
3.1.1 circuit 3
3.2 util 4
3.2.1 lpad 4
3.2.2 opposite-anchor 4
3.2.3 rotate-anchor 4
3.2.4 colors 5
3.3 wire 6
3.3.1 intersection 6
3.3.2 stub 6
3.3.3 wire 7
3.3.4 wire-styles 10
3.4 element 12
3.4.1 elmt 12
3.4.2 alu 14
3.4.3 block 15
3.4.4 extender 16
3.4.5 multiplexer 17
3.4.6 group 18
3.5 gates 20
3.5.1 gate 20
3.5.2 gate-and 22
3.5.3 gate-nand 22
3.5.4 gate-buf 23
3.5.5 gate-not 24
3.5.6 gate-or 24
3.5.7 gate-nor 25
3.5.8 gate-xor 25
3.5.9 gate-xnor 26
3.5.10 iec-gate 27
3.5.11 iec-gate-and 29
3.5.12 iec-gate-nand 29
3.5.13 iec-gate-buf 30
3.5.14 iec-gate-not 30
3.5.15 iec-gate-or 31
3.5.16 iec-gate-nor 32
3.5.17 iec-gate-xor 32
3.5.18 iec-gate-xnor 33
3.6 verilog 34
3.6.1 parse 34
3.6.2 get-module 34
3.6.3 analyze 34
3.6.4 module-ports 35

3.6.5 block 36

1 Introduction

This package provides a way to make beautiful block circuit diagrams using the CeTZ package.

2 Usage

Simply import Circuiteria and call the `circuit` function:

```
#import "@preview/circuiteria:0.2.1"
#circuiteria.circuit({
  import circuiteria: *
  ...
})
```

2.1 Project installation

If you have installed Circuiteria directly in your project, import src/lib.typ and call the `circuit` function:

```
#import "src/lib.typ" as circuiteria
#circuiteria.circuit({
  import circuiteria: *
  ...
})
```

3 Reference

3.1 circuit

- circuit()

3.1.1 circuit

Draws a block circuit diagram

This function is also available at the package root

Parameters

```
circuit(
  body: none array element,
  length: length ratio
) -> none
```

body `none` or `array` or `element`

A code block in which draw functions have been called

length `length` or `ratio`

Optional base unit

Default: `2em`

3.2 util

- lpad()
- opposite-anchor()
- rotate-anchor()

Variables:

- colors

3.2.1 lpad

Pads a string on the left with 0s to the given length

```
#util.lpad("0100", 8)
```

```
00000100
```

Parameters

```
lpad(
  string: str,
  len: int
) -> str
```

string `str`

The string to pad

len `int`

The target length

3.2.2 opposite-anchor

Returns the anchor on the opposite side of the given one

```
#util.opposite-anchor("west")
```

```
east
```

Parameters

```
opposite-anchor(anchor: str) -> str
```

anchor `str`

The input anchor

3.2.3 rotate-anchor

Returns the anchor rotated 90 degrees clockwise relative to the given one

```
#util.rotate-anchor("west")
```

```
north
```

Parameters

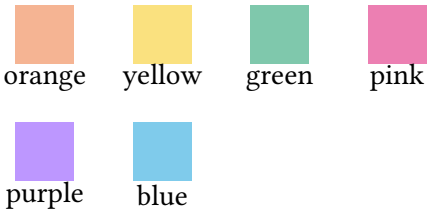
```
rotate-anchor(anchor: str) -> str
```

anchor `str`

The anchor to rotate

3.2.4 colors

Predefined color palette



3.3 wire

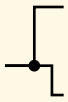
- intersection()
- stub()
- wire()

Variables:

- wire-styles

3.3.1 intersection

Draws a wire intersection at the given anchor



```
wire.wire("w1", ((0, 0), (1, 1)), style: "zigzag")
wire.wire("w2", ((0, 0), (1, -.5)),
          style: "zigzag", zigzag-ratio: 80%)
wire.intersection("w1.zig")
```

Parameters

```
intersection(
  pt: point,
  radius: number,
  fill: color
)
```

pt `point`

A CeTZ compatible point / anchor

radius `number`

The radius of the intersection

Default: `.1`

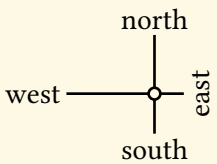
fill `color`

The fill color

Default: `black`

3.3.2 stub

Draws a wire stub (useful for unlinked ports)




```
draw.circle((0, 0), radius: .1, name: "p")
wire.stub("p", "north", name: "north", length: 1)
wire.stub("p", "east", name: "east", vertical: true)
wire.stub("p", "south", name: "south", length: 15pt)
wire.stub("p", "west", name: "west", length: 3em)
```

Parameters

```
stub(
  port-id: str,
  side: str,
  name: none str,
  vertical: bool,
  length: number,
  name-offset: number
)
```

port-id `str`

The port anchor

side `str`

The side on which the port is (one of “north”, “east”, “south”, “west”)

name `none` or `str`

Optional name displayed at the end of the stub

Default: `none`

vertical `bool`

Whether the name should be displayed vertically

Default: `false`

length `number`

The length of the stub

Default: `1em`

name-offset `number`

The name offset, perpendicular to the stub

Default: `0`

3.3.3 wire

Draws a wire between two points

Parameters

```
wire(
  id: str,
  pts: array,
  bus: bool,
  name: none str array,
  name-pos: str,
  slice: none array,
  color: color,
  dashed: bool,
  style: str,
  reverse: bool,
  guided-center: array,
  guided-margins: array,
  guided-sides: array,
  directed: bool,
  rotate-name: bool,
  zigzag-ratio: ratio,
  zigzag-dir: str,
  dodge-y: number,
  dodge-sides: array,
  dodge-margins: array
)
```

id `str`

The wire's id, for future reference (anchors)

pts `array`

The two points (as CeTZ compatible coordinates, i.e. XY, relative positions, ids, etc.)

bus `bool`

Whether the wire is a bus (multiple bits) or a simple signal (single bit)

Default: `false`

name `none` or `str` or `array`

Optional name of the wire. If it is an array, the first name will be put at the start of the wire, and the second at the end

Default: `none`

name-pos `str`

Position of the name. One of: "middle", "start" or "end"

Default: `"middle"`

slice `none` or `array`

Optional bits slice (start and end bit indices). If set, it will be displayed at the start of the wire

Default: `none`

color `color`

The stroke color

Default: `black`

dashed `bool`

Whether the stroke is dashed or not

Default: `false`

style `str`

The wire's style (see `wire-styles` for possible values)

Default: `"direct"`

reverse `bool`

If true, the start and end points will be swapped (useful in cases where the start point depends on the end point, for example with perpendiculars)

Default: `false`

guided-center `array`

the horizontal and vertical guides of the center guides (only with style "guided")

Default: `(50%, 50%)`

guided-margins `array`

the start and end of guided margins of the wire (only with style "guided")

Default: `(5%, 5%)`

guided-sides `array`

the side of start and end array (must be either "north", "south", "west", "east") (only work with style "guided")

Default: `("east", "west")`

directed `bool`

If true, the wire will be directed, meaning an arrow will be drawn at the endpoint

Default: `false`

rotate-name `bool`

If true, the name will be rotated according to the wire's slope

Default: `true`

zigzag-ratio `ratio`

Position of the zigzag vertical relative to the horizontal span (only with style "zigzag")

Default: `50%`

zigzag-dir `str`

The zigzag's direction. As either "vertical" or "horizontal" (only with dstyle "zigzag")

Default: `"vertical"`

dodge-y `number`

Y position to dodge the wire to (only with style "dodge")

Default: `0`

dodge-sides `array`

The start and end sides (going out of the connected element) of the wire (only with style "dodge")

Default: `("east", "west")`

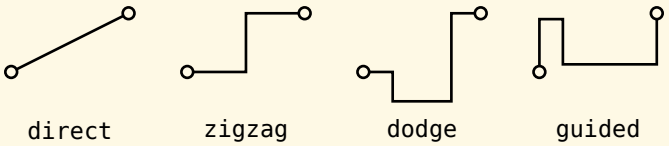
dodge-margins `array`

The start and end margins (i.e. space before dodging) of the wire (only with style "dodge")

Default: `(5%, 5%)`

3.3.4 wire-styles

List of valid wire styles




```
for i in range(4) {
  draw.circle((i * 3, 0), radius: .1, name: "p" + str(i * 2))
  draw.circle((i * 3 + 2, 1), radius: .1, name: "p" + str(i * 2 + 1))
  draw.content((i * 3 + 1, -1), raw(wire.wire-styles.at(i)))
}
wire.wire("w1", ("p0", "p1"), style: "direct")
wire.wire("w2", ("p2", "p3"), style: "zigzag")
wire.wire("w3", ("p4", "p5"), style: "dodge",
          dodge-y: -0.5, dodge-margins: (0.5, 0.5))
wire.wire("w4", ("p6","p7"), style: "guided",
          guided-center:(20%, 40%), guided-margins: (90%,87%),
          guided-sides: ("north","south"))
```

3.4 element

- elmt()
- alu()
- block()
- extender()
- multiplexer()
- group()

3.4.1 elmt

Draws an element

Parameters

```
elmt(
  draw-shape: function,
  x: number dictionary,
  y: number dictionary,
  w: number,
  h: number,
  name: none str,
  name-anchor: str,
  ports: dictionary,
  ports-margins: dictionary,
  fill: none color,
  stroke: stroke,
  id: str,
  auto-ports: bool,
  ports-y: dictionary,
  debug: dictionary
)
```

draw-shape `function`

Draw function

Default: `default-draw-shape`

x `number` or `dictionary`

The x position (bottom-left corner).

If it is a dictionary, it should be in the format `(rel: number, to: str)`, where `rel` is the offset and `to` the base anchor

Default: none

y `number` or `dictionary`

The y position (bottom-left corner).

If it is a dictionary, it should be in the format `(from: str, to: str)`, where `from` is the base anchor and `to` is the id of the port to align with the anchor

Default: none

w `number`

Width of the element

Default: none

h `number`

Height of the element

Default: none

name `none` or `str`

Optional name of the block

Default: none

name-anchor `str`

Anchor for the optional name

Default: `"center"`

ports `dictionary`

Dictionary of ports. The keys are cardinal directions (“north”, “east”, “south” and/or “west”). The values are arrays of ports (dictionaries) with the following fields:

- `id` (`str`): (Required) Port id
- `name` (`str`): Optional name displayed **in** the block
- `clock` (`bool`): Whether it is a clock port (triangle symbol)
- `invert` (`bool`): Whether it is an active-low port (bubble symbol)
- `vertical` (`bool`): Whether the name should be drawn vertically

Default: `(:)`

ports-margins `dictionary`

Dictionary of ports margins (used with automatic port placement). They keys are cardinal directions (“north”, “east”, “south”, “west”). The values are tuples of (`<start>`, `<end>`) margins (numbers)

Default: `(:)`

fill `none` or `color`

Fill color

Default: none

stroke `stroke`

Border stroke

Default: `black + 1pt`

id `str`

The block id (for future reference)

Default: `""`

auto-ports `bool`

Whether to use auto port placements or not. If false, `draw-shape` is responsible for adding the appropiate ports

Default: `true`

ports-y `dictionary`

Dictionary of the ports y offsets (used with `auto-ports: false`)

Default: `(:)`

debug `dictionary`

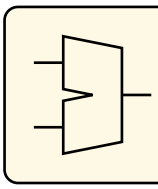
Dictionary of debug options.

Supported fields include:
- `ports`: if true, shows dots on all ports of the element

Default:
```
(
  ports: false
)
```

3.4.2 alu

Draws an ALU with two inputs



```
element.alu(x: 0, y: 0, w: 1, h: 2, id: "alu")
wire.stub("alu-port-in1", "west")
wire.stub("alu-port-in2", "west")
wire.stub("alu-port-out", "east")
```

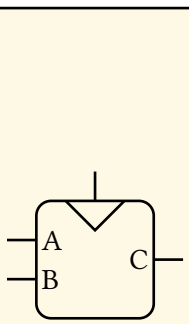
For parameters description, see `elmt()`

Parameters

```
alu(
  x,
  y,
  w,
  h,
  name,
  name-anchor,
  fill,
  stroke,
  id,
  debug
)
```

3.4.3 block

Draws a block element




```
element.block(
  x: 0, y: 0, w: 2, h: 2, id: "block",
  ports: (
    north: ((id: "clk", clock: true),),
    west: ((id: "in1", name: "A"),
           (id: "in2", name: "B")),
    east: ((id: "out", name: "C"),)
  )
)
wire.stub("block-port-clk", "north")
wire.stub("block-port-in1", "west")
wire.stub("block-port-in2", "west")
wire.stub("block-port-out", "east")
```

For other parameters description, see `elmt()`

Parameters

```
block(
  x,
  y,
  w,
  h,
  name,
  name-anchor,
  ports,
  ports-margins,
  fill,
  radius: number length ratio dictionary,
  stroke,
  id,
  debug
)
```

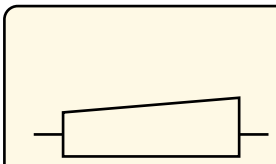
radius `number` or `length` or `ratio` or `dictionary`

The corner radius of the block. See CeTZ documentation for more information

Default: `0.5em`

3.4.4 extender

Draws a bit extender



```
element.extender(
  x: 0, y: 0, w: 3, h: 1,
  id: "extender"
)
wire.stub("extender-port-in", "west")
wire.stub("extender-port-out", "east")
```

For other parameters description, see `elmt()`

Parameters

```
extender(
  x,
  y,
  w,
  h,
  name,
  name-anchor,
  fill,
  stroke,
  id,
  h-ratio: ratio,
  align-out: bool,
  debug
)
```

h-ratio `ratio`

The height ratio of the left side relative to the full height

Default: `75%`

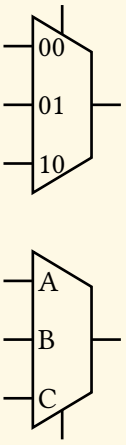
align-out `bool`

If true, the output and input ports are aligned, otherwise, the output port is centered on the right side

Default: `true`

3.4.5 multiplexer

Draws a multiplexer




```
element.multiplexer(
  x: 0, y: 0, w: 1, h: 3,
  id: "multiplexer",
  entries: 3
)
wire.stub("multiplexer.north", "north")
wire.stub("multiplexer-port-out", "east")

element.multiplexer(
  x: 0, y: -4, w: 1, h: 3,
  id: "multiplexer2",
  entries: ("A", "B", "C")
)
wire.stub("multiplexer2.south", "south")
wire.stub("multiplexer2-port-out", "east")

for i in range(3) {
  wire.stub("multiplexer-port-in" + str(i), "west")
  wire.stub("multiplexer2-port-in" + str(i), "west")
}
```

For other parameters description, see `elmt()`

Parameters

```
multiplexer(
  x,
  y,
  w,
  h,
  name,
  name-anchor,
  entries: int array,
  h-ratio: ratio,
  fill,
  stroke,
  id,
  debug
)
```

entries `int` or `array`

If it is an integer, it defines the number of input ports (automatically named with their binary index). If it is an array of strings, it defines the name of each input.

Default: 2

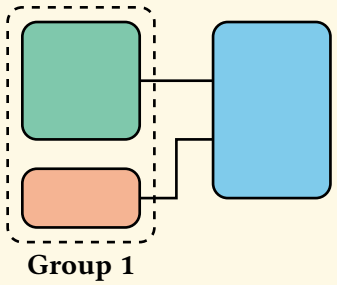
h-ratio `ratio`

The height ratio of the right side relative to the full height

Default: `60%`

3.4.6 group

Draws a group of elements




```
element.group(
  id: "g1", name: "Group 1", stroke: (dash: "dashed"),
  {
    element.block(id: "b1", w: 2, h: 2,
      x: 0, y: 1.5,
      ports: (east: ((id: "out"),)),
      fill: util.colors.green
    )
    element.block(id: "b2", w: 2, h: 1,
      x: 0, y: 0,
      ports: (east: ((id: "out"),)),
      fill: util.colors.orange
    )
  }
)
element.block(id: "b3", w: 2, h: 3,
  x: (rel: 1, to: "g1.east"),
  y: (from: "b1-port-out", to: "in1"),
  ports: (west: ((id: "in1"), (id: "in2"))),
  fill: util.colors.blue
)
wire.wire("w1", ("b1-port-out", "b3-port-in1"))
wire.wire("w2", ("b2-port-out", "b3-port-in2"),
          style: "zigzag")
```

Parameters

```
group(
  body: elements function,
  id: str,
  name: str,
  name-anchor: str,
  fill: color,
  stroke: stroke,
  padding: float length array dictionary,
  radius: number
)
```

body `elements` or `function`

Elements to group

id `str`

see `elmt()`

Default: `""`

name `str`

The group's name

Default: `none`

name-anchor `str`

The anchor for the name.
Note: the name will be placed on the **outside** of the group

Default: `"south"`

fill `color`

see `elmt()`

Default: `none`

stroke `stroke`

see `elmt()`

Default: `black + 1pt`

padding `float` or `length` or `array` or `dictionary`

The inside padding:
- float / length: same for all sides
- array: either (<all>,), (<vertical>, <horizontal>) or (<top>, <right>, <bottom>, <left>)
- dictionary: valid keys are "top", "right", "bottom" and "left"

Default: `0.5em`

radius `number`

The corner radius

Default: `0.5em`

3.5 gates

- gate()
- gate-and()
- gate-nand()
- gate-buf()
- gate-not()
- gate-or()
- gate-nor()
- gate-xor()
- gate-xnor()
- iec-gate()
- iec-gate-and()
- iec-gate-nand()
- iec-gate-buf()
- iec-gate-not()
- iec-gate-or()
- iec-gate-nor()
- iec-gate-xor()
- iec-gate-xnor()

3.5.1 gate

Draws a logic gate. This function is also available as `element.gate()`

Parameters

```
gate(
  draw-shape: function,
  x: number dictionary,
  y: number dictionary,
  w: number,
  h: number,
  inputs: int,
  fill: none color,
  stroke: stroke,
  id: str,
  inverted: str array,
  inverted-radius: number,
  debug: dictionary
)
```

draw-shape `function`

see `elmt()`

Default: `default-draw-shape`

x `number` or `dictionary`

see `elmt()`

Default: `none`

y `number` or `dictionary`

see `elmt()`

Default: `none`

w `number`

see `elmt()`

Default: `none`

h `number`

see `elmt()`

Default: `none`

inputs `int`

The number of inputs

Default: `2`

fill `none` or `color`

see `elmt()`

Default: `none`

stroke `stroke`

see `elmt()`

Default: `black + 1pt`

id `str`

see `elmt()`

Default: `""`

inverted `str` or `array`

Either “all” or an array of port ids to display as inverted

Default: `()`

inverted-radius `number`

The radius of inverted ports dot

Default: `0.1`

debug `dictionary`

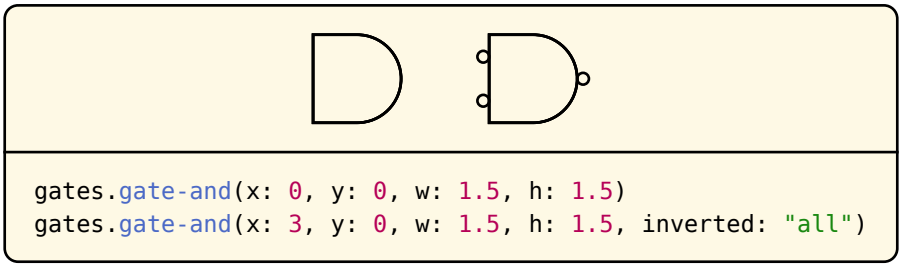
see `elmt()`

Default: `(`
```
    ports: false
  )
```

3.5.2 gate-and

Draws an AND gate. This function is also available as `element.gate-and()`

For parameters, see `gate()`



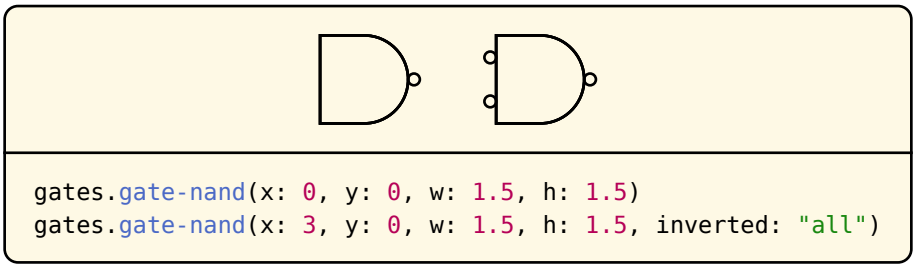
```
gates.gate-and(x: 0, y: 0, w: 1.5, h: 1.5)
gates.gate-and(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

Parameters
```
gate-and(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.3 gate-nand

Draws an NAND gate. This function is also available as `element.gate-nand()`

For parameters, see `gate()`



```
gates.gate-nand(x: 0, y: 0, w: 1.5, h: 1.5)
gates.gate-nand(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

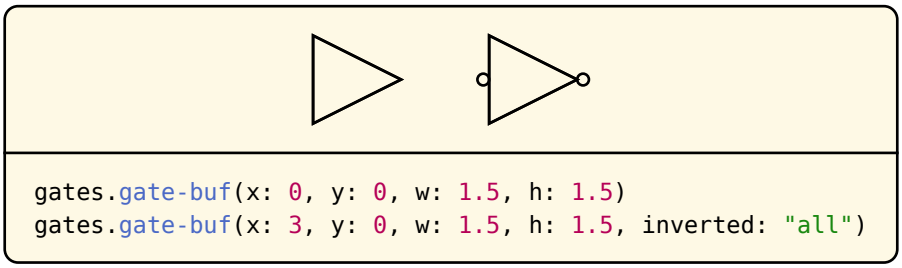
Parameters

```
gate-nand(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.4 gate-buf

Draws a buffer gate. This function is also available as `element.gate-buf()`

For parameters, see `gate()`



```
gates.gate-buf(x: 0, y: 0, w: 1.5, h: 1.5)
gates.gate-buf(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

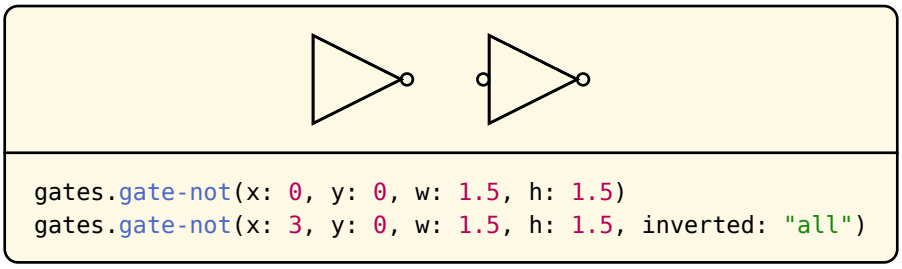
Parameters

```
gate-buf(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.5 gate-not

Draws a NOT gate. This function is also available as `element.gate-not()`

For parameters, see `gate()`



```
gates.gate-not(x: 0, y: 0, w: 1.5, h: 1.5)
gates.gate-not(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

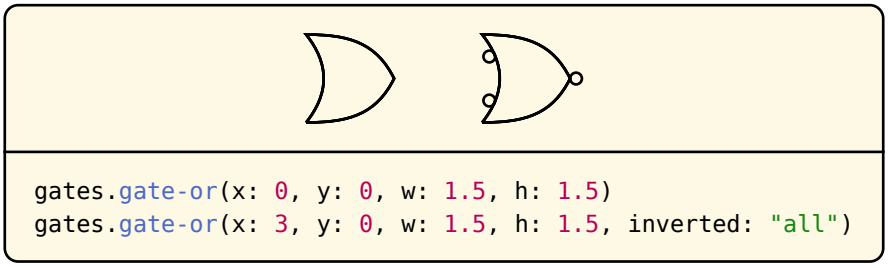
Parameters

```
gate-not(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.6 gate-or

Draws an OR gate. This function is also available as `element.gate-or()`

For parameters, see `gate()`



```
gates.gate-or(x: 0, y: 0, w: 1.5, h: 1.5)
gates.gate-or(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

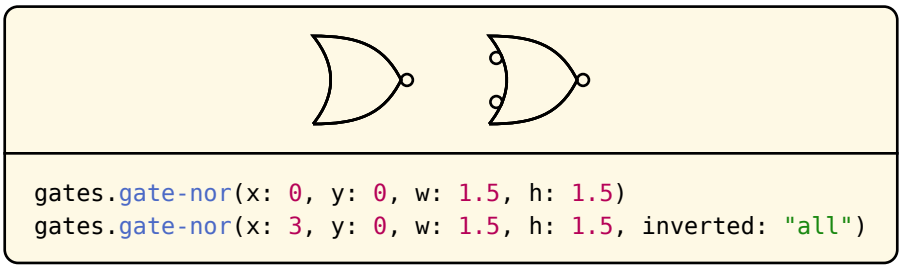
Parameters

```
gate-or(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.7 gate-nor

Draws a NOR gate. This function is also available as `element.gate-nor()`

For parameters, see `gate()`



```
gates.gate-nor(x: 0, y: 0, w: 1.5, h: 1.5)
gates.gate-nor(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

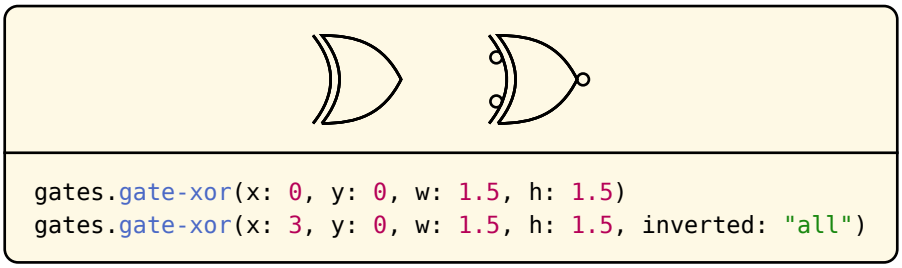
Parameters

```
gate-nor(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.8 gate-xor

Draws a XOR gate. This function is also available as `element.gate-xor()`

For parameters, see `gate()`



```
gates.gate-xor(x: 0, y: 0, w: 1.5, h: 1.5)
gates.gate-xor(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

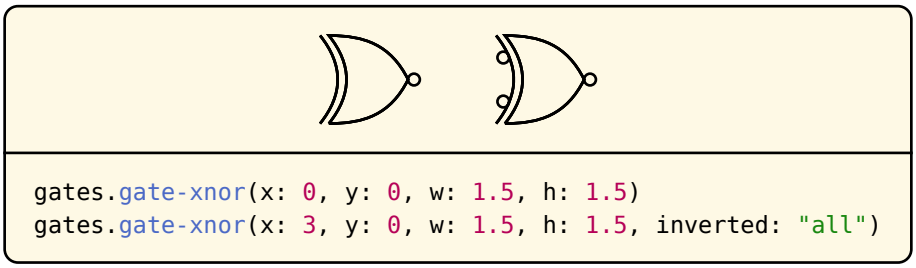
Parameters

```
gate-xor(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.9 gate-xnor

Draws a XNOR gate. This function is also available as `element.gate-xnor()`

For parameters, see `gate()`



```
gates.gate-xnor(x: 0, y: 0, w: 1.5, h: 1.5)
gates.gate-xnor(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

Parameters

```
gate-xnor(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.10 iec-gate

Draws a logic gate. This function is also available as `element.iec-gate()`

Parameters

```
iec-gate(
  draw-shape: function,
  x: number dictionary,
  y: number dictionary,
  w: number,
  h: number,
  inputs: int,
  fill: none color,
  stroke: stroke,
  id: str,
  inverted: str array,
  inverted-radius: number,
  debug: dictionary,
  symbol: str
)
```

draw-shape `function`

see `elmt()`

Default: `default-draw-shape`

x `number` or `dictionary`

see `elmt()`

Default: `none`

y `number` or `dictionary`

see `elmt()`

Default: `none`

w `number`

see `elmt()`

Default: `none`

h `number`

see `elmt()`

Default: `none`

inputs `int`

The number of inputs

Default: `2`

fill `none` or `color`

see `elmt()`

Default: `none`

stroke `stroke`

see `elmt()`

Default: `black + 1pt`

id `str`

see `elmt()`

Default: `""`

inverted `str` or `array`

Either “all” or an array of port ids to display as inverted

Default: `()`

inverted-radius `number`

The radius of inverted ports dot

Default: `0.1`

debug `dictionary`

see `elmt()`

Default:
```
(
    ports: false
)
```

symbol `str`

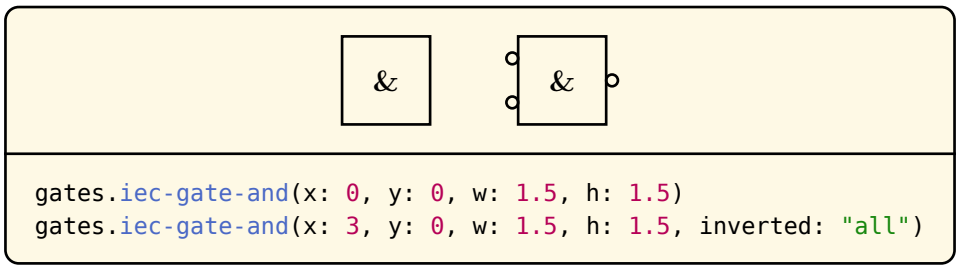
The symbol to display at the center of the gate

Default: `""`

3.5.11 iec-gate-and

Draws an IEC-AND gate. This function is also available as `element.iec-gate-and()`

For parameters, see `iec-gate()`




```
gates.iec-gate-and(x: 0, y: 0, w: 1.5, h: 1.5)
gates.iec-gate-and(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

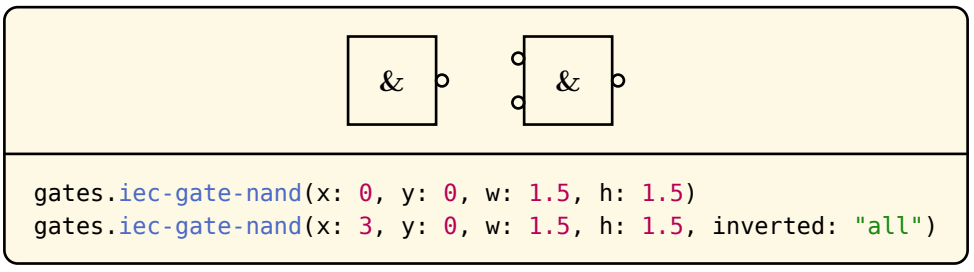
Parameters

```
iec-gate-and(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.12 iec-gate-nand

Draws an IEC-NAND gate. This function is also available as `element.iec-gate-nand()`

For parameters, see `iec-gate()`




```
gates.iec-gate-nand(x: 0, y: 0, w: 1.5, h: 1.5)
gates.iec-gate-nand(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

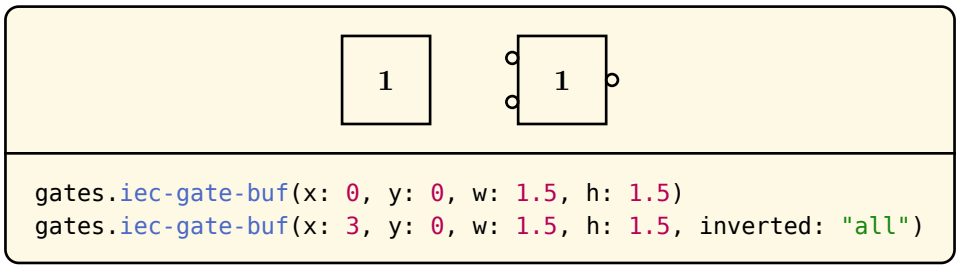
Parameters

```
iec-gate-nand(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.13 iec-gate-buf

Draws an IEC buffer gate. This function is also available as `element.iec-gate-buf()`

For parameters, see `iec-gate()`



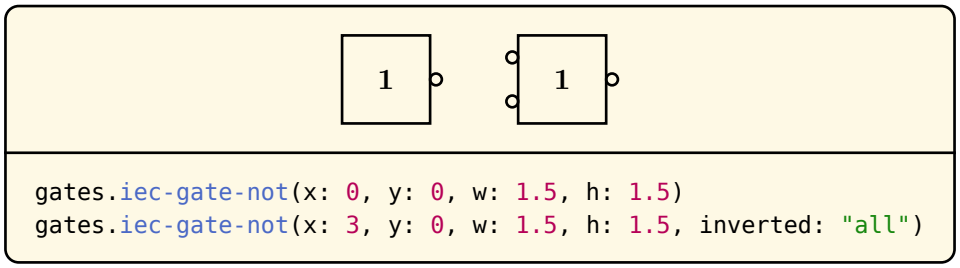
Parameters

```
iec-gate-buf(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.14 iec-gate-not

Draws an IEC NOT gate. This function is also available as `element.iec-gate-not()`

For parameters, see `iec-gate()`




```
gates.iec-gate-not(x: 0, y: 0, w: 1.5, h: 1.5)
gates.iec-gate-not(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

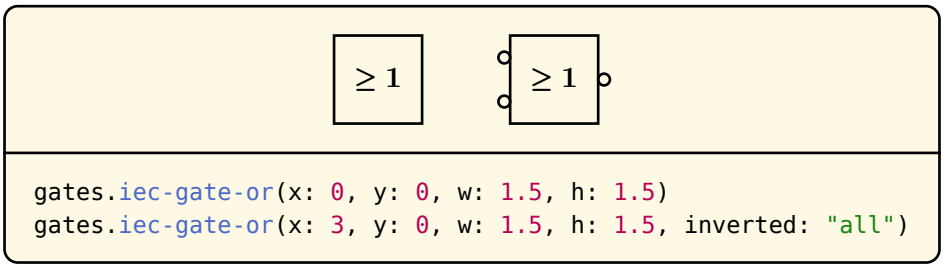
Parameters

```
iec-gate-not(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.15 iec-gate-or

Draws an IEC-OR gate. This function is also available as `element.iec-gate-or()`

For parameters, see `iec-gate()`




```
gates.iec-gate-or(x: 0, y: 0, w: 1.5, h: 1.5)
gates.iec-gate-or(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

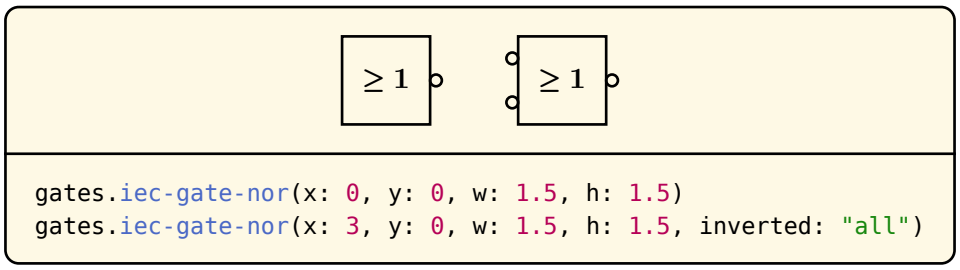
Parameters

```
iec-gate-or(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.16 iec-gate-nor

Draws an IEC-NOR gate. This function is also available as `element.iec-gate-nor()`

For parameters, see `iec-gate()`




```
gates.iec-gate-nor(x: 0, y: 0, w: 1.5, h: 1.5)
gates.iec-gate-nor(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

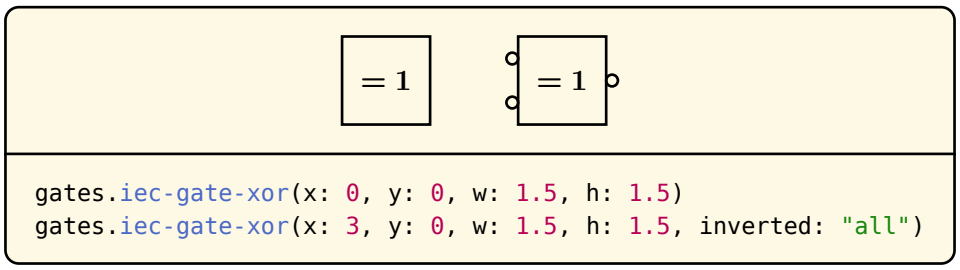
Parameters

```
iec-gate-nor(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.17 iec-gate-xor

Draws an IEC-XOR gate. This function is also available as `element.iec-gate-xor()`

For parameters, see `iec-gate()`




```
gates.iec-gate-xor(x: 0, y: 0, w: 1.5, h: 1.5)
gates.iec-gate-xor(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

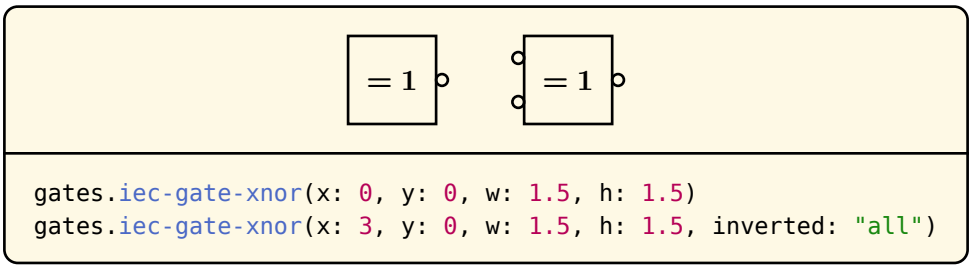
Parameters

```
iec-gate-xor(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.5.18 iec-gate-xnor

Draws an IEC-XNOR gate. This function is also available as `element.iec-gate-xnor()`

For parameters, see `iec-gate()`




```
gates.iec-gate-xnor(x: 0, y: 0, w: 1.5, h: 1.5)
gates.iec-gate-xnor(x: 3, y: 0, w: 1.5, h: 1.5, inverted: "all")
```

Parameters

```
iec-gate-xnor(
  x,
  y,
  w,
  h,
  inputs,
  fill,
  stroke,
  id,
  inverted,
  debug
)
```

3.6 verilog

- parse()
- get-module()
- analyze()
- module-ports()
- block()

3.6.1 parse

Parses Verilog/SystemVerilog source code and returns the declared modules.

Both ANSI (`module m(input clk, ...)`) and non-ANSI (`module m(clk, ...); input clk;`) port styles are supported.

```
#let modules = verilog.parse(read("my_module.v"))
```

Parameters

```
parse(source: str) -> array
```

source `str`

The Verilog source code, typically obtained with `read()`

3.6.2 get-module

Returns a single module from Verilog source code.

Parameters

```
get-module(
  source: str array dictionary,
  name: none str
) -> dictionary
```

source `str` or `array` or `dictionary`

Verilog source code, an array of parsed modules or an already parsed module

name `none` or `str`

Name of the wanted module. If `none`, the first module is returned

Default: `none`

3.6.3 analyze

Analyses the ports of a module and infers AXI4, AXI4-Lite and AXI4-Stream buses as well as clock and reset signals.

The returned dictionary contains:

- `buses`: array of inferred buses. Each bus has the fields `kind` (`"axi"`, `"axi-lite"` or `"axi-stream"`), `prefix`, `role` (`"master"` or `"slave"`), `dir`, `data-width`, `addr-width` and `signals`
- `clocks`: array of clock ports
- `resets`: array of reset ports (with an additional `active-low` field)
- `signals`: array of the remaining ports

Parameters

```
analyze(
  module: str array dictionary,
  name: none str
) -> dictionary
```

module `str` or `array` or `dictionary`

Verilog source code or a parsed module

name `none` or `str`

Name of the wanted module (when several are present)

Default: `none`

3.6.4 module-ports

Builds the ports dictionary of a Verilog module, ready to be given to `elmt()`.

Inputs (and slave buses) are placed on the west side, outputs (and master buses) on the east side, clocks and resets on top of the west side.

Parameters

```
module-ports(
  module: str array dictionary,
  name: none str,
  show-widths: bool,
  show-bus-info: bool,
  clock-side: str,
  reset-side: str
) -> dictionary
```

module `str` or `array` or `dictionary`

Verilog source code or a parsed module

name `none` or `str`

Name of the wanted module (when several are present)

Default: `none`

show-widths `bool`

Whether to append the signal widths to their names

Default: `true`

show-bus-info `bool`

Whether to display the bus kind and data width below the bus name

Default: `false`

clock-side `str`

Side on which clock ports are placed

Default: `"west"`

reset-side `str`

Side on which reset ports are placed

Default: `"west"`

3.6.5 block

Draws a block from a Verilog module description.

AXI4, AXI4-Lite and AXI4-Stream signals are grouped into single bus ports, and clock/reset signals are automatically detected and grouped together.

```
#circuiteria.circuit({
  circuiteria.verilog.block(read("axi_dma.v"), x: 0, y: 0)
})
```

The port ids are the signal names (or the bus prefixes), so wires can be attached with `"<id>-port-<signal>"`.

Parameters

```
block(
  module: str array dictionary,
  module-name: none str,
  x: number dictionary,
  y: number dictionary,
  w: auto number,
  h: auto number,
  id: auto str,
  name: auto none str content,
  name-anchor: str,
  show-widths: bool,
  show-bus-info: bool,
  stubs: bool,
  stub-length: number length,
  clock-side: str,
  reset-side: str,
  fill: none color,
  stroke: stroke,
  radius: number length ratio dictionary,
  ports-margins: dictionary,
  debug: dictionary
)
```

module `str` or `array` or `dictionary`

Verilog source code or a parsed module

module-name `none` or `str`

Name of the wanted module (when several are present)

Default: `none`

x `number` or `dictionary`

The x position (bottom-left corner)

Default: `0`

y `number` or `dictionary`

The y position (bottom-left corner)

Default: `0`

w `auto` or `number`

Width of the block. Computed from the port names if `auto`

Default: `auto`

h `auto` or `number`

Height of the block. Computed from the port count if `auto`

Default: `auto`

id `auto` or `str`

The block id. Defaults to the module name

Default: `auto`

name `auto` or `none` or `str` or `content`

Name displayed in the block. Defaults to the module name

Default: `auto`

name-anchor `str`

Anchor for the displayed name

Default: `"south"`

show-widths `bool`

Whether to append the signal widths to their names

Default: `true`

show-bus-info `bool`

Whether to display the bus kind and data width below the bus name

Default: `false`

stubs `bool`

Whether to draw an unnamed wire stub on every port

Default: `true`

stub-length `number` or `length`

Length of the automatic stubs

Default: `1em`

clock-side `str`

Side on which clock ports are placed

Default: `"west"`

reset-side `str`

Side on which reset ports are placed

Default: `"west"`

fill `none` or `color`

Fill color

Default: `none`

stroke `stroke`

Border stroke

Default: `black + 1pt`

radius `number` or `length` or `ratio` or `dictionary`

The corner radius of the block

Default: `0.5em`

ports-margins `dictionary`

Dictionary of ports margins

Default: `(:)`

debug `dictionary`

Dictionary of debug options

Default: `(ports: false)`